

What the Survey Missed — presenter guide

One HTML file, offline, no AI at runtime. You need: a screen to project or share; no other materials.

Learning goals

Not stated objectives — what the activity itself demonstrably shows, one per stage:

1. A fixed round returns a detection history for every site, not a count of animals — that is the only kind of evidence this investigation produces.
2. A difference in detections can come from occupancy, from detection probability, or from both — one round cannot tell you which.
3. More fixed rounds improve the evidence, but a finite realization can still disagree with the true occupancy ranking.
4. The habitat with more detections is not always the habitat with more occupied sites — and one visit alone cannot separate occupancy from detection probability.

Where this sits on the path

The tenth of ten AI-built classroom tools in Part four, and the last demo in the tour. Like the four "investigation" demos before it (wavelet, ion flight, inhibitor, missing time), it teaches a discipline through the same grammar: predict, run, then explain before reveal. Its own discipline is ecology — survey design under imperfect detection — and its assumptions are one tap away, the same standard every earlier demo in the tour is held to.

One question for the session

Can fewer sightings mean fewer occupied sites? Not necessarily — a detection difference between two habitats can come from a difference in occupancy, a difference in how easy the species is to detect, or both, and one fixed round of surveying cannot tell you which.

*The one sentence to land: **a detection is not an animal, and a difference in detections is not by itself a difference in occupancy.** In this scenario, the habitat with the higher true occupancy (Habitat A, $\psi = 0.8$) is EXPECTED to show fewer detections after one round than the habitat with lower occupancy but higher detectability (Habitat B, $\psi = 0.5, p = 0.8$).*

The shape of the session

STAGE	WHAT HAPPENS	TIME
1 — Survey	Predict a habitat, then run the one fixed round across all 48 sites.	2 min
2 — Interpret	Choose the claim actually justified by the round-1 detection difference.	2–3 min
3 — Revisit	Optionally repeat the fixed survey with rounds 2 and 3.	2 min
4 — Compare	Answer the transfer question, then reveal the synthetic model and, optionally, the likelihood view.	3 min

Stage 1 — Survey

Question. When you survey every one of 48 fixed sites once, what have you actually measured?

Predict prompt. Guess which habitat, A or B, will show more detections after this first round — before pressing Survey round 1.

Activity. Have the room commit to a habitat by show of hands before you press **Survey round 1 · 48 site-visits**. The maps fill in live; ask what a colored site means before revealing that it is only a detection, not a confirmed absence of the

species elsewhere.

Expected observation. With the declared values (A: $\psi = 0.8$, $p = 0.25$; B: $\psi = 0.5$, $p = 0.8$), Habitat B is EXPECTED to show more detections despite having the LOWER occupancy probability — the room's habit of predicting the higher-occupancy habitat to "win" the detection count is the misconception this stage exists to surface.

Check-yourself answer key. "Does more detections mean more occupied sites?" — No: a detection difference can come from occupancy, detection probability, or both.

Teaching limit. The generating ψ and p values are invented to make this one point; they are not measured from a real survey and should not be read as typical field values.

Misconception: more detections means more animals were found.

A detection history counts site-visits where the species was seen, not individuals. A habitat can produce more detections while having fewer occupied sites, if its detection probability is higher.

Stage 2 — Interpret

Question. Is a detection difference between habitats, by itself, enough to conclude an occupancy difference?

Predict prompt. A quick yes/no vote before the three options: does a detection difference alone prove an occupancy difference?

Activity. Read the round-1 recap aloud, then have the room choose among the three claims. Only the cautious claim ("could reflect occupancy, detection probability, or both") unlocks the rest of the demo; the other two are genuine, plausible-sounding overreaches, not strawmen.

Expected observation. The cautious claim is the only one the model actually supports; "the habitat with fewer detections must have lower occupancy" is exactly backwards for the declared scenario.

Check-yourself answer key. "What should you look for before trusting an occupancy ranking from one round?" — Evidence about detection probability, not only the raw detection count.

Teaching limit. This stage tests recognition of the correct claim; it does not by itself estimate ψ or p from data — that is what the optional likelihood view in Stage 4 illustrates, and even it cannot separate the two from one visit.

Misconception: a detection difference already settles which habitat has more occupied sites.

It settles nothing about occupancy alone. The same detection difference is equally consistent with an occupancy difference, a detection-probability difference, or a mix of both.

Stage 3 — Revisit

Question. Does adding a round change which habitat shows the higher observed fraction, and does it have to?

Predict prompt. Before running round 2, ask whether the room thinks the round-1 ranking can flip.

Activity. Run round 2, then round 3 if time allows. Watch the observed fractions in the metric cards move toward (but not exactly reach) their theoretical expectations from the Details drawer.

Expected observation. The observed fractions get closer to their $K = 3$ expectations (A 0.4625, B 0.496) but a real seeded run can still land either side of the expected ranking — the exact outcome the room sees will vary by which scenario is loaded.

Check-yourself answer key. "Does enough rounds guarantee the ranking is correct?" — No: more rounds improve the evidence but do not guarantee it, because the two expected fractions stay close.

Teaching limit. Only three rounds are modeled; the demo does not claim what happens at ten or a hundred rounds, where the law of large numbers would narrow this gap further.

Misconception: enough repeated surveys always produce the "correct" ranking.

More visits improve the evidence but a finite sample can still disagree with the underlying model, especially when the two habitats' expected values are close, as they are here by $K = 3$.

Stage 4 — Compare

Question. Which habitat actually has more occupied sites in this seeded scenario — the one with more detections, or the other one?

Predict prompt. Before revealing, guess which habitat has the higher REALIZED occupancy, not the higher detected fraction.

Activity. Answer the transfer question (a colleague's overreach) as a room, then press **Reveal synthetic map and model parameters**. Point out the newly visible "occupied but missed" sites — they were occupied the whole time and simply were not detected in the rounds run. If time allows, open the optional likelihood view and note that it never produces a single confident answer from one visit.

Expected observation. Habitat A (the higher-occupancy habitat) reveals MORE occupied sites despite having shown fewer detections throughout — the inversion the whole demo has been building toward.

Check-yourself answer key. "What does 'occupied but missed' show about detection?" — A truly occupied site can produce zero detections across every round run, because detection probability is less than one.

Teaching limit. The likelihood view uses only observed histories from one chosen habitat at a time and a coarse 11-by-11 grid; it illustrates the ψ/p ambiguity, it does not fit a model.

Misconception: revealing the map is the "real" measurement and the survey rounds were only practice.

The reveal is not a measurement either — it is the generating truth of a synthetic scenario, shown for comparison. In a real field study, no such reveal ever happens; the survey rounds ARE the only evidence a researcher gets.

Discussion questions

1. If you could only run one more round in ONE habitat, which would you choose to best distinguish "Habitat A has lower detection probability" from "Habitat A has lower occupancy"? Why does a single extra round in either habitat still leave you short of a clean answer?
2. The reveal shows Habitat A has more occupied sites but fewer detections. Name one real-world reason a species might be more occupant-rich but harder to detect (habitat structure, behavior, timing).
3. The optional likelihood view uses only ONE habitat's histories at a time. What would combining both habitats' histories into one estimate assume that this demo does not?
4. This demo never estimates ψ or p with a number and a confidence interval, the way RPresence or a similar tool would. What would a real analysis need that this demo's single seeded scenario cannot supply?

Presenter controls

The stage tabs switch stages; Interpret, Revisit and Compare unlock as the room progresses and re-lock on Reset.

Details (in each stage's intro) opens that stage's assumptions and sources. **Guide** reopens the orientation card;

Settings holds **Open Presenter Notes** (this document, on screen), **Presentation mode** (scales the page for a projector), **Reset** (the whole demo back to first load, in place — no page reload), and two scenario controls: **Replay as practice** (same seed, clears progress) and **New seeded scenario** (draws the next one). Site squares in either map are clickable at any time to read that one site's own detection history.

Sources and attribution

MacKenzie, D. I., et al. (2002), "Estimating Site Occupancy Rates When Detection Probabilities Are Less Than One," *Ecology*, 83(8), 2248–2255. U.S. Geological Survey: [Estimating site occupancy rates when detection probabilities are less than one](#). This is the source context for separating occupancy probability and detection probability, and for the detection-history likelihood structure this model implements.

Donovan, T. M., Hines, J. E., & MacKenzie, D. I. (2024), [occupancyTuts: Occupancy modelling tutorials with RPresence](#), U.S. Geological Survey. Current USGS teaching materials on occupancy/detection-history workflows with RPresence, cited here for occupancy-modeling teaching context. This replaces a 2023 RPresence workshop page previously cited at mbr-pwrc.usgs.gov/workshops/occupancy2023online.html, which as of a 2026-09-15 fleet-wide citation sweep now blanket-redirects (confirmed against a second path on the same subdomain) to a generic USGS center homepage with no occupancy-modeling content.

Neither source establishes a current Murray State assignment or endorsement.

What the Survey Missed · Bryant Harrison · Murray State University · Offline synthetic teaching model. This HTML file is the canonical guide source: the printable sheet, the PDF, and the in-app presenter notes all come from it.